

Evaluating and Equivalent ExpressionsQuestion Bank 3.06 **ANSWER KEY***Show your work. Circle your final answer.***For questions 1 through 5, apply the laws of exponents to evaluate. Show your steps.**

1. $\frac{0.25^{-2} \cdot 0.25^5}{0.25^3}$

$$\frac{0.25^3}{0.25^3} = 0.25^0 = 1$$

2. $(-3)^3 \cdot \left(\frac{-5}{3}\right)^{-4}$

$$(-3)^3 \cdot \left(\frac{-3}{5}\right)^4 = \frac{(-3)^7}{5^4} = -\frac{2187}{625}$$

3. $(3^3)^{-3}$

$$3^{-9} = \frac{1}{3^9} = \frac{1}{19,683}$$

4. $\left(\frac{48}{6}\right)^{-2}$

$$8^{-2} = \frac{1}{64} \quad (\text{or} \quad \left(\frac{6}{48}\right)^2 = \frac{36}{2304} = \frac{1}{64})$$

5. $(3^4 \cdot 2^{-4})^{-1}$

$$3^{-4} \cdot 2^4 = \frac{2^4}{3^4} = \frac{16}{81}$$

Evaluate each expression in questions 6 through 10. Then name the laws you used.

6. $\frac{6^{-2} \cdot 6^5}{6}$

36 — product of powers, quotient of powers

7. $\frac{(-9)^4}{(-9)^0 \cdot (-9)^3}$

-9 — zero exponent, quotient of powers

$$(-9)^1.$$

8. $5^2 \cdot \left(\frac{1}{4}\right)^{-1}$

25 · 4 = 100 — negative exponent

9. $\frac{(-2.5)^0 \cdot (-2.5)^1}{(-2.5)^{-2}}$

$$(-2.5)^3 = -15.625 \quad \text{— zero exponent, quotient of powers}$$

10. $\left(\frac{7}{5}\right)^{-4}$

$$\left(\frac{5}{7}\right)^4 = \frac{625}{2401} \quad \text{— negative exponent, power of a quotient}$$

Common bases.

11. Write each expression as a single power with the base given.

a. 8^3 with base 2

$$2^9$$

b. 25^3 with base 5

$$5^6$$

c. 64^2 with base 8

$$8^4$$

d. $2^4 \cdot 4^2$ with base 2

$$2^8$$

$$4^2 = 2^4.$$

12. Marquise says $\left(\frac{1}{8}\right)^3 \cdot 2^{-3}$ and $\left(\frac{1}{4}\right)^5 \cdot 2^{-2}$ are equivalent because both can be written as 2^{-12} .

a. Write the first expression as a single power of 2. Show your work.

$$2^{-9} \cdot 2^{-3} = 2^{-12}$$

$$(1/8)^3 = (2^{-3})^3 = 2^{-9}.$$

b. Write the second expression as a single power of 2. Show your work.

$$2^{-10} \cdot 2^{-2} = 2^{-12}$$

$$(1/4)^5 = (2^{-2})^5 = 2^{-10}.$$

c. Is Marquise correct?

Yes — both are 2^{-12} .

13. Which expression is equivalent to $\frac{2^6}{3^3}$?

● A. $3^{-3} \cdot 2^6$

○ B. $\left(\frac{2}{3}\right)^3$

○ C. $\left(\frac{2}{3}\right)^2$

○ D. 6^3

A

$3^{-3} \cdot 2^6$ moves the 3^3 down; the value is $64/27$.

14. Find the value of n that makes each statement true.

a. $8^n = 2^{12}$

$n = 4$

$(2^3)^n = 2^{3n}$.

b. $\left(\frac{1}{9}\right)^n = 3^6$

$n = -3$

$(3^{-2})^n = 3^{-2n} = 3^6$.